

Supporting agroforestry innovations with holistic, decision-focused modelling: case studies

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Agroforestry systems are complex and therefore difficult to model. Attempts at developing comprehensive system models have usually produced complicated, data-hungry frameworks that are difficult to apply, especially where the respective agroforestry system is not yet in place. Such models have struggled to include the host of institutional, biophysical and economic barriers that agroforestry pioneers are confronted with. These shortcomings have limited the potential of agroforestry models to support real-life decisions on agroforestry innovations. We present a novel approach to supporting agroforestry decisions, based on the principles of decision analysis, an interdisciplinary research methodology that aims to provide the best possible advice to decision-makers based on the current state of knowledge. Decision analysis is designed to include all important aspects of a decision and all relevant uncertainties, using all available sources of information. It is capable of comprehensively covering the entire context, within which decisions on complex systems are made. Here we present the results of three recent decision analysis models that supported agroforestry decisions, on agroforestry adoption by smallholder farmers in Vietnam, on the viability of earning carbon credits through an agroforestry scheme in Costa Rica and on new agroforestry initiatives in Germany. In all cases, decision models were developed through participatory modeling, and probabilistic simulations of the relative merits of each agroforestry option were elaborated. When comprehensively evaluating the multiple benefits of agroforestry, tree-based systems usually appeared preferable to systems without trees. However, decision analysis often identified critical knowledge gaps related, for instance, to the value of particular benefits or to farmers' time preference. Based on such guidance, critical knowledge limitations can be

pinpointed and addressed through further research. While decision analysis only generates coarse forecasts, these are often sufficient for clearly identifying preferable courses of action, offering actionable advice to decision-makers considering the adoption of innovative agroforestry options.